TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology
Wardha Road, Nagpur - 441108
Accredited with NAAC A+ Grade
Approved by AICTE, New Delhi, Govt. of Maharashtra

(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)
Department of Information Technology
Session 2025-26(Even Semester)

VISION	MISSION
To emerge as a learning hub and center of excellence in the domain of Information Technology	[M1] To impart quality technical education through effective teaching learning process. [M2] To provide a platform to address societal issues as well as challenges faced by IT industries. [M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT. [M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning.

Class Test (CT)-2

Semester: VI

Programme: B. Tech Information Technology

Course: Natural Language Processing

Time: 1 Hr 30 min **Max Marks:** 30 Marks

Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

BIT33601	Course Outcome
CO1	Classify the ability to preprocess text data using NLP techniques.
CO2	Demonstrate syntactic parsing methods, including context-free grammars and dependency parsing
CO3	Analyze semantic structures and apply information retrieval techniques to extract meaningful insights from text.
CO4	Interpret language models and perform text classification using algorithms in NLP.
CO5	Evaluate real-world applications of Natural Language Processing

Group A				
Que. 1	Answer the following Question	CO	BTL	Marks
a.	Explain semantic network in NLP	CO3	2	2
b.	What problem of RNN is solved by LSTM	CO4	2	2
c.	Illustrate trends in NLP	CO5	2	2

Group B				
Que. 2	Answer the following Question (Solve any Two)	CO	BTL	Marks

a.	Explain different approaches to Word Sense Disambiguation (WSD) with examples.	CO3	2	4
b.	A search engine must rank document based on user query. Explain how vector space model works in this case.	CO3	4	4
c.	Discuss semantic roles and semantic networks in NLP and highlight the differences between them.	C03	2	4
Que 3.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Classify the applications of NLP	CO4	2	4
b.	Explain N-gram language models (unigram, bigram, trigram) with suitable examples.	CO4	2	4
c.	Describe the four types of Machine Learning approaches used in NLP.	CO4	2	4
Que 4.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	An online company wants to use Natural Language Processing to analyze customer reviews, translate content into multiple languages, and provide automatic responses to customers. Explain how different NLP applications can be used for this purpose.	CO5	2	4
b.	Discuss ethical consideration in NLP.	CO5	2	4
c.	Outline Machine Translation and its different approaches with examples	CO5	2	4

Bloom's Taxonomy Level (BTL)- Remember , Understand , Apply, Analyze , Evaluate , Create